Ethan J. Moger

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EDUCATION

Cornell University | Duffield College of Engineering **Ithaca, NY**
Major: *Mechanical Engineering* | ***Cumulative GPA:*** *4.264* *Aug 2024 – May 2028*
Relevant Coursework: Dynamics, Statics, Thermodynamics, Differential Eqs, Linear Algebra, Physics I/II, Calculus I/II/III

Queensbury Senior High School **Queensbury, NY**
NYS Advanced Regents Diploma with Honors, Seal of Distinction in Math & Science *Sep 2020 – Jun 2024*
International Baccalaureate Diploma | **Class Rank:** 1 / 241 | **Cumulative GPA:** 99.320 / 100.000

WORK & RELEVANT EXPERIENCE

Fluor Marine Propulsion (Naval Nuclear Laboratory) **Niskayuna, NY**
Incoming Mechanical Engineering Intern – Nuclear Reactor Equipment Design *May 2027 – Aug 2027*

X-Ray Optical Systems **East Greenbush, NY**
Engineering Intern *May 2026 – Aug 2026*
- Redesigned a replacement for an obsolete part in SolidWorks, reducing cost by 18% while meeting all design constraints
- Updated SolidWorks/PDM assembly files across 7 product lines, improving inventory management and documentation
- Verified regulatory compliance for ~900 parts and subassemblies, organizing records using Epicor and Excel

Cornell University Hyperloop Project Team **Ithaca, NY**
Magnetics Subteam Lead, Mechanical Engineer *Sep 2025 – Present*
- Directed magnetics development and coordinated cross-functional subsystem design, integration, and implementation
- Utilized Ansys Maxwell to design and optimize a linear induction motor, doubling thrust output versus prior designs
- Simplified Maxwell simulations by redesigning coil models and utilizing stranded conductors, reducing mesh complexity

Mechanical Design Project: Spotted Lanternfly Control **Ithaca, NY**
Cornell University MAE 2250 Student *Jan 2026 – May 2026*
- Designed and prototyped a handheld device to remove spotted lanternfly egg masses, with a capacity of 100 egg masses
- Developed CAD models in Fusion 360 and fabricated components through 3D printing, machining, and assembly
- Conducted testing and analysis, achieving 2.65s egg mass removal time and earning an Outstanding Design nomination

Glove Dryer Design & Provisional Patent **Queensbury, NY**
Member/Contributor *Sep 2022 – Jun 2023*
- Developed a soccer goalie glove-drying system through research and design with a focus on material preservation
- Designed and 3D printed the product in OnShape, incorporating feedback and advice from engineering professionals
- Presented the prototype to a panel of engineers and later applied for and was granted a provisional patent

LEADERSHIP EXPERIENCE

Physics 1112 / Physics 2213 Learning Assistant **Ithaca, NY**
Undergraduate TA *Aug 2025 – May 2026*
- Facilitated collaborative problem-solving in discussion sessions to support peer learning in calculus-based physics
- Enhanced communication and teaching skills through weekly seminars on physics education theory and reflective practice
- Partnered with faculty to promote inclusive instruction and provide feedback to improve course delivery and engagement

Adirondack Solar Alliance **Queensbury, NY**
Co-Founder *Dec 2021 – May 2023*
- Co-founded a not-for-profit organization that has raised over $100,000 to install solar panels in Adjuntas, Puerto Rico
- Assembled and gave formal presentations to local businesses, securing financial support for past and future projects
- Developed the donation page, coordinated technology-related tasks for fundraising, and participated in the first installation

SKILLS, ACTIVITIES, AWARDS & INTERESTS

Skills: SolidWorks, Ansys Maxwell, Fusion 360, MATLAB, OnShape, Python, Microsoft Office, Collaboration, Problem Solving
Activities: American Society of Mechanical Engineers, Intramural Soccer (Captain), Doubles Pickleball League
Awards: President's Award for Educational Excellence, RIT Science and Math Scholarship, AP Scholar
Interests: 3D Printing/Design, Programming, Mountain Biking, Skiing, Pick-Up Soccer, Pickleball, Cliff Jumping, Piano